

Intro to Spatial Epidemiology (PUBH7632 – Semester 2, 2026)

Session 2: Is the pattern more than just visual (beyond random chance)?

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ODeSI

Operational Research and Decision
Support for Infectious Diseases

M3 hub

Modelling, Maths
and Maps



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Acknowledgement of Country

- The University of Queensland (UQ) acknowledges the Traditional Owners and their custodianship of the lands on which we meet.
- We pay our respects to their Ancestors and their descendants, who continue cultural and spiritual connections to Country.
- We recognise their valuable contributions to Australian and global society.

*The Brisbane River pattern from A Guidance Through Time
by Casey Coolwell and Kyra Mancktelow.*



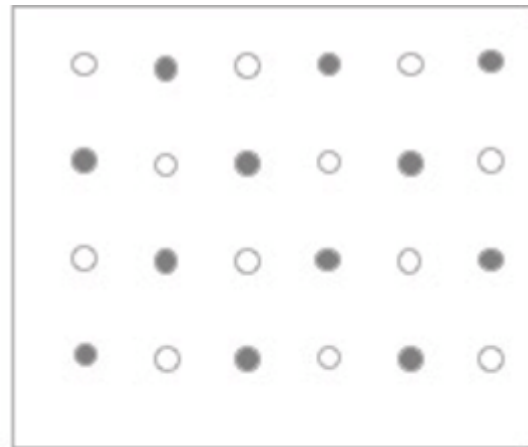
Overview of spatial epidemiology module

Session	Focus	Lecture and tutorial outcomes
1. From health data to maps (21 Sep 2026)	Foundations of spatial epidemiology and GIS: when place changes a health question; point, areal, raster, and network data; layers, attributes, joins, CRS, scale, and resolution. Introduces counts, crude and age-standardised rates, risk, confidentiality, denominators, MAUP, ecological fallacy, and uncertainty.	QGIS tutorial: import boundaries and health/population data; check IDs, missingness, and CRS; join data; calculate crude rates; create and critique choropleth maps under alternative classifications.
2. Is the pattern more than visual? (7 October 2026)	Exploratory spatial analysis: spatial dependence and heterogeneity; neighbourhood definitions; spatial weights; global Moran's I ; local Moran's I ; Getis–Ord hotspots; multiple testing and sensitivity analysis.	R tutorial: create alternative spatial-weights matrices; calculate and interpret global Moran's I ; map local clusters/hotspots; compare results under different neighbourhood definitions; distinguish a statistical signal from causal evidence.
3. Mapping underlying risk (12 October 2026)	Disease mapping beyond raw rates: small-number instability; observed versus expected cases; SIR/SMR; non-spatial and spatial empirical Bayes smoothing; exploratory hotspots versus cluster detection and scan statistics.	R tutorial: calculate expected counts and standardised ratios; compare crude rates, standardised measures, and smoothed estimates; identify changed interpretations; conduct an introductory scan-statistic exercise and interpret clusters as surveillance signals.
4. From description to explanation (19 October 2026)	Moving from “where?” to “why?”: epidemiological questions and causal assumptions; count regression with population offsets; residual spatial autocorrelation; spatial models and spatial random effects; confounding, uncertainty, ecological inference, and decision communication.	R tutorial: fit an area-level count model; assess residual spatial autocorrelation; compare non-spatial and spatial estimates; communicate evidence, a recommended public-health action, and one important caveat.

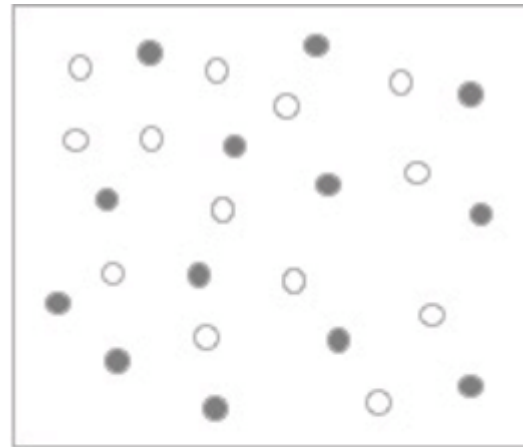
Lecture aims and content

- Introduce spatial dependence versus spatial heterogeneity.
- Explain neighbours and spatial weights matrices: contiguity versus distance-based definitions.
- Global versus local questions:
 - Global Moran's : Is there overall spatial autocorrelation?
 - Local Moran's and Getis-Ord : Where are unusually similar values or hotspots?

Exploratory analysis of discrete data



Negative spatial
autocorrelation
(or uniform distribution)



Spatial randomness
(or no correlation)



Positive spatial
autocorrelation
(clustered)

Spatial Autocorrelation

- Classical statistical models often assume independent observations
- In health data, observations may be related across space and time
- Tobler's First Law:
"All things are related, but nearby things are more related than distant things."
- Spatial autocorrelation: nearby areas tend to have more similar—or more dissimilar—values than expected by chance

Global Measures and Local Measures of Spatial Autocorrelation

- **Global Measures**

- One statistic to summarise pattern in whole study area
- Helps describe clustering and homogeneity

- **Local Measures**

- Location-specific statistics (one value calculated per unit area)
- Helps identify cluster/hotspots and heterogeneity

To summarise: three main categories of tests to assess spatial clustering/autocorrelation

1. Global: a single test for general patterns and spatial autocorrelation over the entire study region or area
 - Moran's I
 - Geary's C
2. Local: search for specific regions or areas where clustering is observed above or below expected levels
 - Examples: areas with high or low crime rates
 - Local Moran's I
 - Getis-Ord G_i^*
 - Spatial Scan Statistic (e.g. Kulldorff)
3. Focal: specialised statistics search only in regions around fixed locations
 - Example: cancers around nuclear reactors/power plants
 - Stone's Test
 - Bithell
 - Tango

We do not cover these statistics in this module but just to be aware.

Measures of spatial autocorrelation

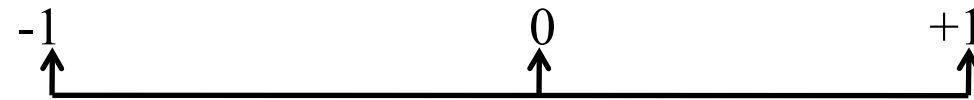
- Measures of Spatial Autocorrelation appropriate for discrete areal data:
 - Anselin Moran's I
 - Geary's C
 - Getis-Ord G statistic
- In this lecture and in the subsequent tutorial we will focus on the Moran's I statistic (both global and local)

Are attribute values in neighbouring polygons (suitably weighted) more similar than is expected by chance?

- **Distance-based approach:** represent each polygon by its centroid and examine how similarity changes with distance using a **variogram** or **spatial correlogram**.
- **Neighbour-based approach:** define a set of neighbours and their weights, then use **Moran's I** to quantify overall spatial autocorrelation.
- **Key idea:** nearby areas can be given greater influence than distant areas, for example using inverse-distance weights

Moran's I

- The most common measure of spatial autocorrelation
- Interpretation similar to a correlation coefficient i.e. ranges a scale from -1 to $+1$

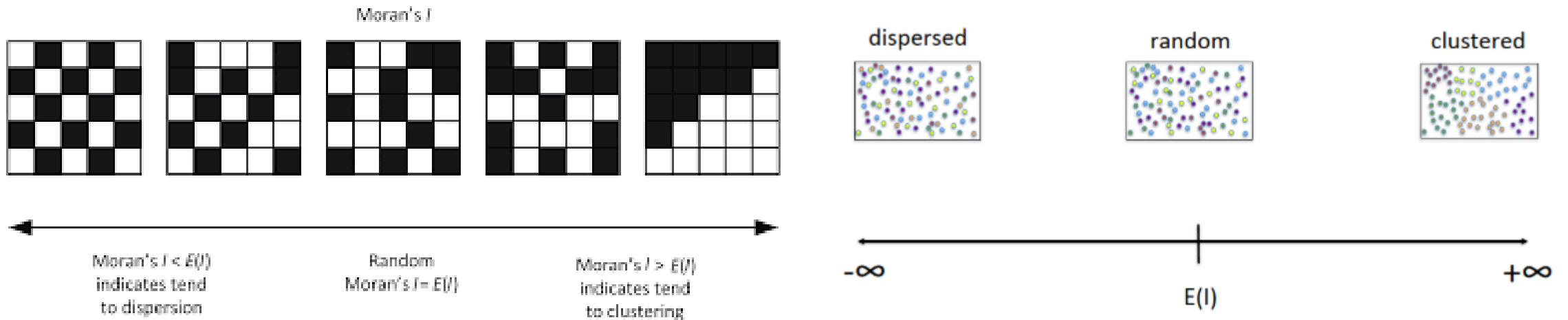


high negative spatial
autocorrelation

no spatial
autocorrelation*

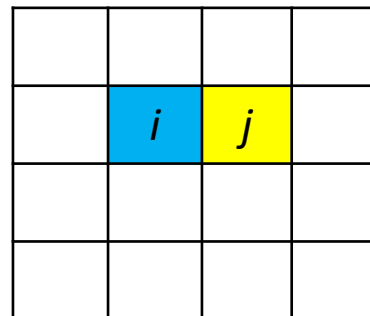
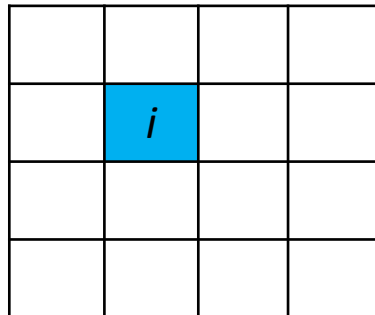
high positive spatial
autocorrelation

Can also use it as an index for dispersion/random/cluster patterns.

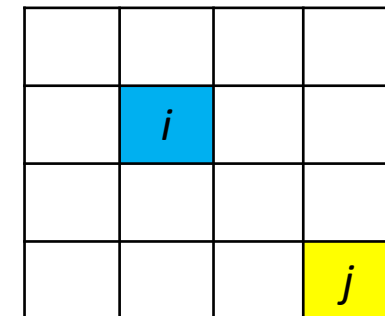


Defining neighbours: proximity matrices

- Generally, neighbours are defined as sharing a common boundary or vertices
- This can be expressed formally as a proximity matrix
 - For two areas i and j , neighbourhood weighting is W_{ij}
 - $W_{ij} = 1$ if i and j share a common boundary; 0 if they don't
 - Alternatively, W_{ij} could be distance between centroids of areas i and j , or = 1 if j is one of the K nearest neighbours of i



$$W_{ij} = 1$$

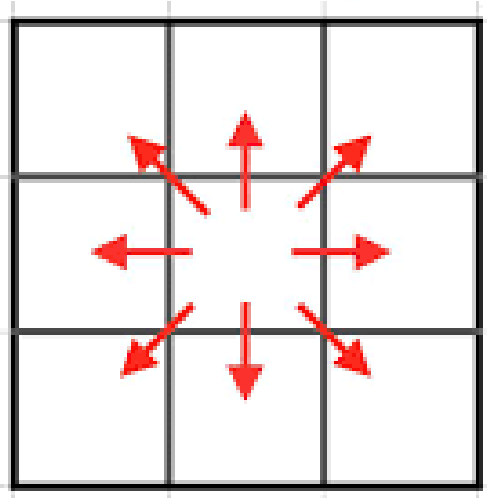


$$W_{ij} = 0$$

Spatial neighbors based on adjacency

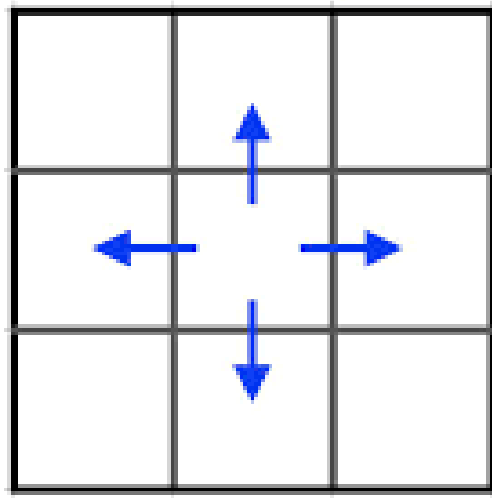
Regular/lattice:

QUEEN contiguity



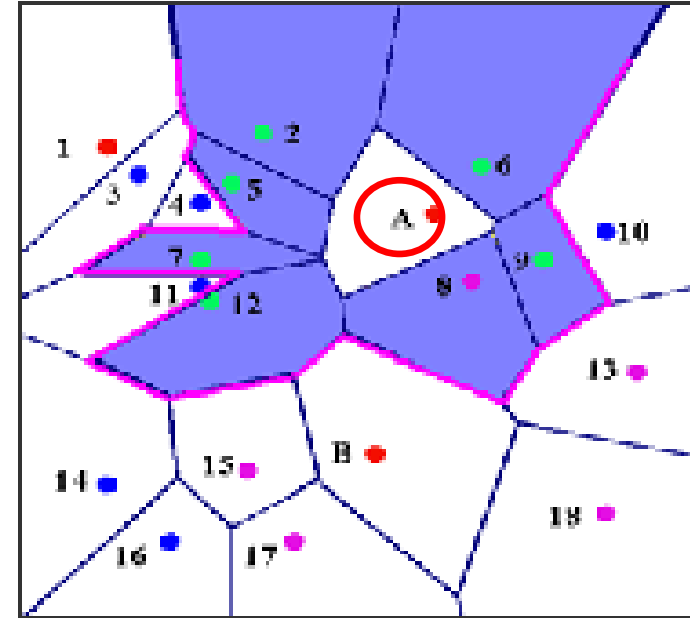
sharing a boundary
or point

ROOK contiguity



sharing a boundary

Irregular:



What type of contiguity is the
above an example of based
on unit of interest highlight
with red circle?

1st, 2nd and k-order adjacency matrices

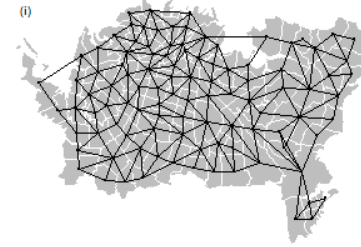
Using about **10–15 nearby locations*** is usually enough for the model to detect the main spatial pattern. Including more neighbours may add computing time, but often produces little improvement in results

Distance based centroids within 1000 and 2500 miles for example

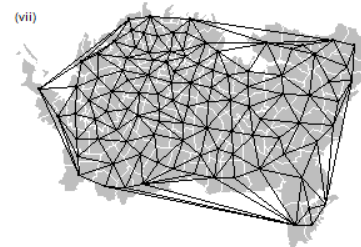
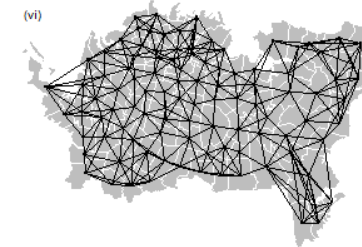
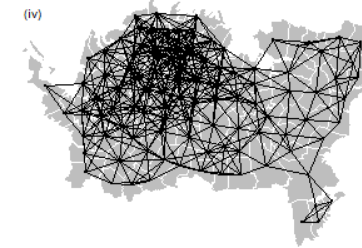
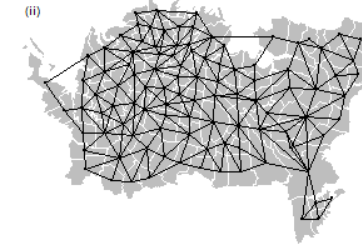
K-Nearest neighbours
3 and 6 for example

Delaunay triangulation

1st order Rook adjacency



1st order Queen adjacency



2nd order Rook adjacency

*Datta, S. Banerjee, A. O. Finley, and A. E. Gelfand. Hierarchical nearest-neighbor gaussian process models for large geostatistical datasets. Journal of the Americal Statistical Association, 2016.

Global Moran's I

- Compares the value of variable at any one location (in this case discrete area) with the value at all other locations (or discrete areas)

Number of objects $\rightarrow n$

$$I = \frac{\sum_i \sum_j w_{ij} (y_i - \bar{y})(y_j - \bar{y})}{\sum_i \sum_j w_{ij} \sum_i (y_i - \bar{y})^2}$$

Sum of weights

Variance

Large positive if both y_i and y_j are larger or smaller than the mean
Large negative if one larger than mean and the other is smaller than the mean
Around 0 (null) if one or especially both are close to the mean

- Where:
 - y_i is variable at a particular location
 - y_j is variable at another location
 - $\bar{y}$ is the mean across the entire area
 - W_{ij} is weight applied to comparison between i and j

Relative location: e.g. adjacency matrix
Absolute location: e.g. inverse distance from centroid

- Varies between +1 (positive spatial correlation) and -1 (negative spatial correlation)
- What does a test statistic value of 0 imply?**

Modification of Moran's I for point data

- Moran's I

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \frac{\sum_i \sum_j w_{ij} (y_i - \bar{y})(y_j - \bar{y})}{\sum_i (y_i - \bar{y})^2}$$

- Modification for point data

- Replace weights matrix with distance bands, width h
- Pre-normalise Y values by subtracting means
- Count number of other points in each band, N(h)

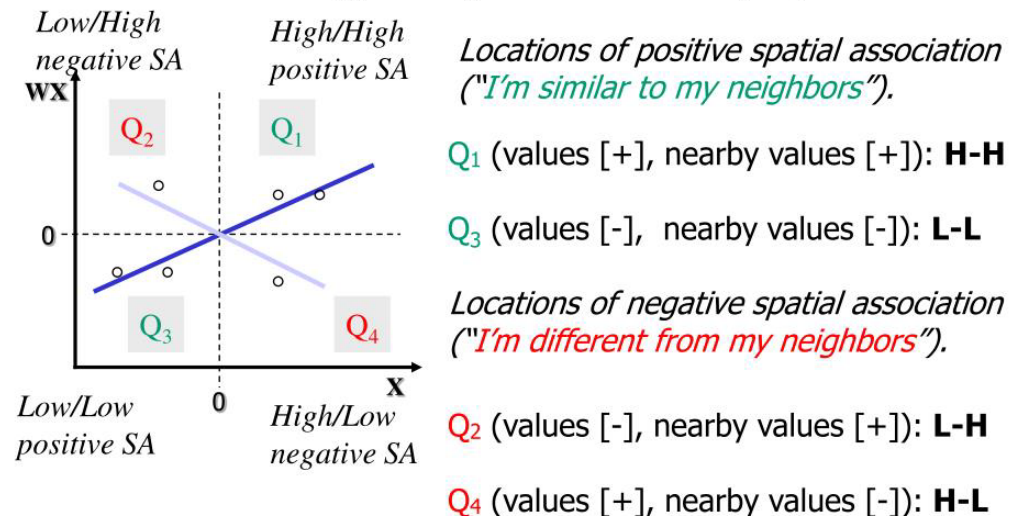
$$I(h) = N(h) \frac{\sum_i \sum_j y_i y_j}{\sum_i y_i^2}$$

Morans I scatter plot

- Moran's I can be interpreted as the correlation between variable, X , and the “spatial lag” of X formed by averaging all the values of X for the neighboring polygons
- We can then draw a scatter diagram between these two variables (in standardized form): X and $\text{lag-}X$ (or W_X)

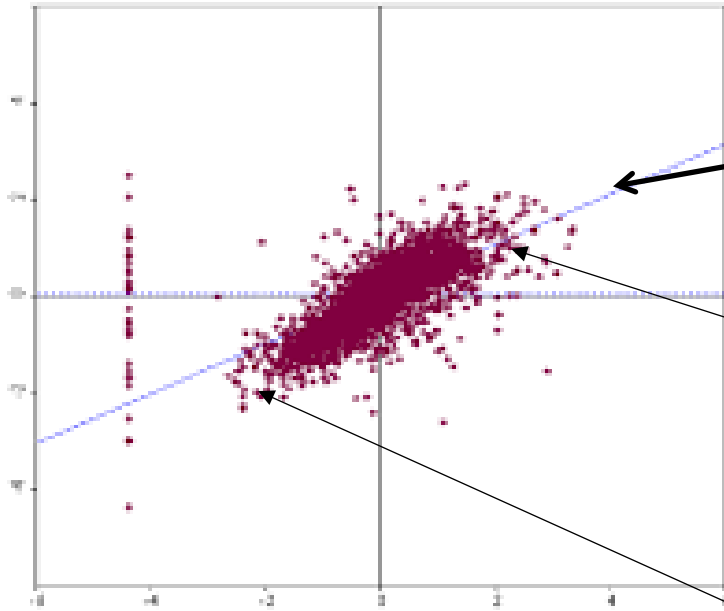
Quadrants of Moran Scatterplot

Each quadrant corresponds to one of the four different types of spatial association (SA)



Example Moran's Scatter Plot

Avg. Neighbors HDI Score

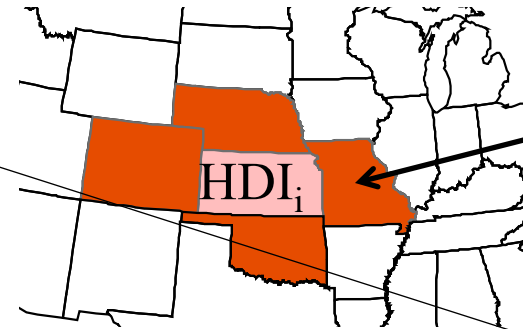


Least squares "best fit" line to the points.
The slope of this *regression line* is Moran's I

HDI Score

Scaled so values
are expressed as
the number of
standard
deviations away
from the overall
mean

The Moran's I statistic of 0.5175,
significant at the $p < 0.001$



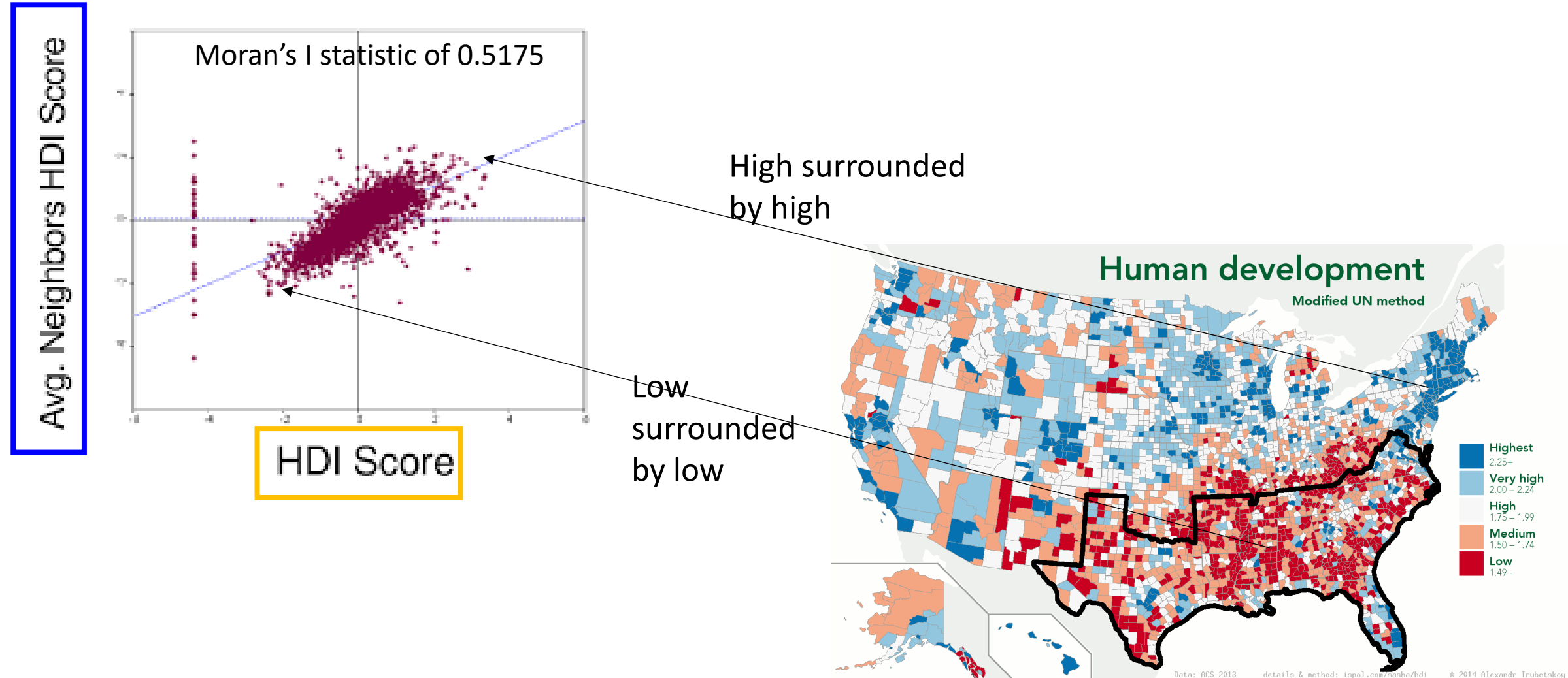
Lag HDI_i
is average
of these

High values
surrounded by high
values

Low values
surrounded by low
values

Example Moran's Scatter Plot

- Scatterplot of (scaled) **Y** vs. (scaled) **Lag-Y**



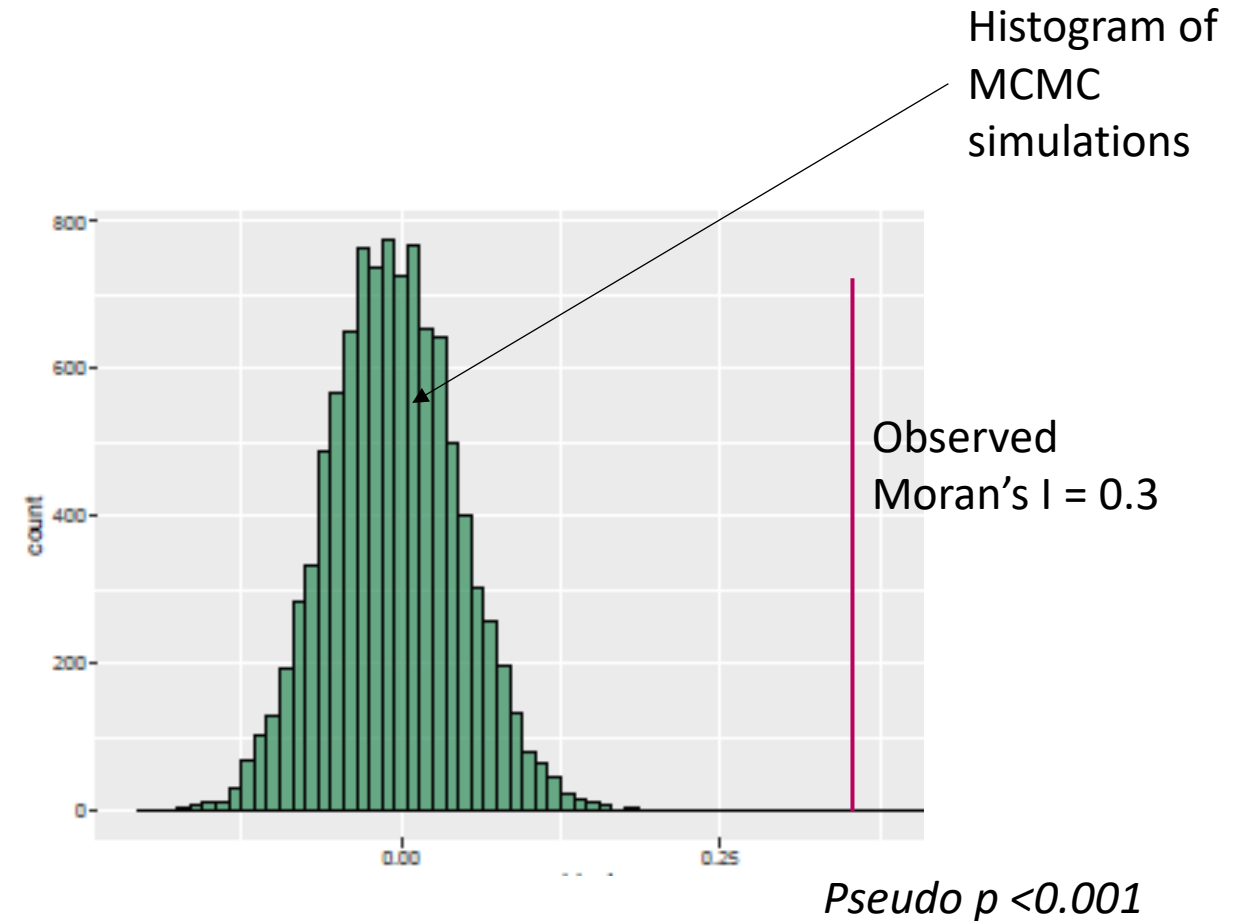
Hypothesis testing for Moran's I – is what we're seeing of statistical significance?

- **Null hypothesis = complete spatial randomness**
- We can generate a null hypothesis for spatial data using Monte Carlo simulation – ie randomly re-assign our set of outcomes across all of our spatial units, and recalculating the Moran's I statistic *a lot of times* (e.g. 10,000)
- Compare the distribution of the MCMC estimates to the Moran's I from the real data to calculate a pseudo *p* value:

$$(N_x + 1)/(N + 1)$$

Where: N_x = # simulated values above original estimate

N = # simulations



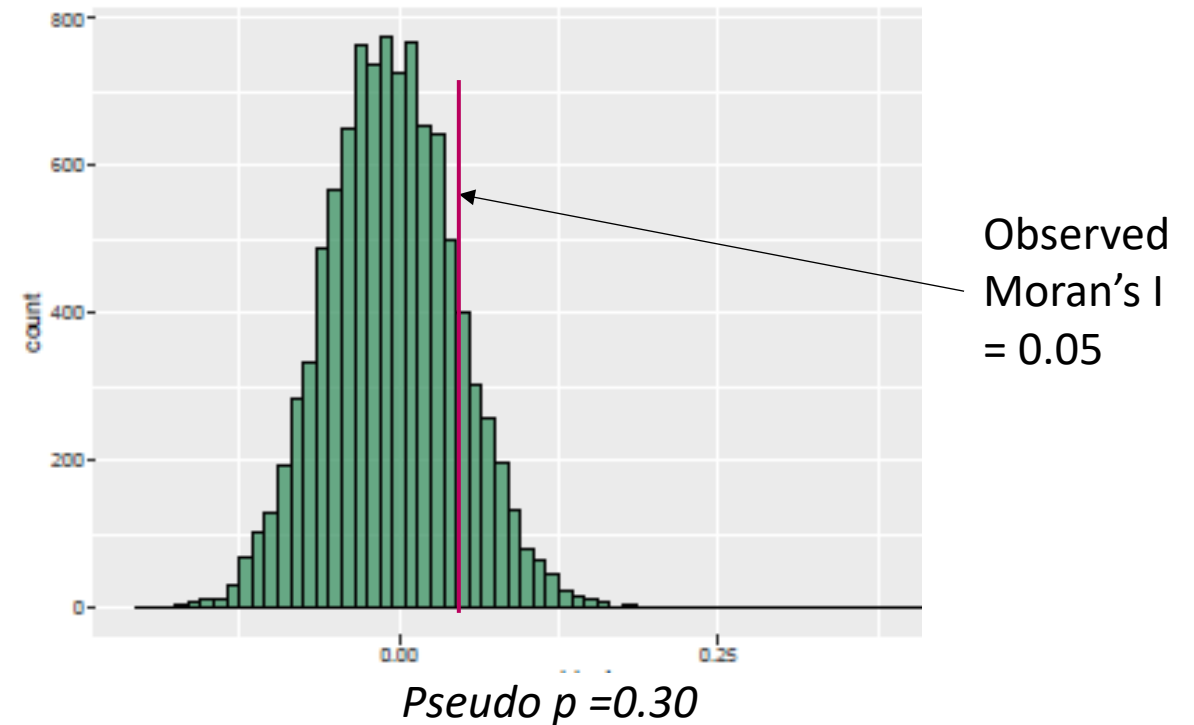
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Global Measures and Local Measures

- **Global Measures**

- One statistic to summarise pattern in whole study area
- Helps describe clustering and homogeneity

- **Local Measures**

- Location-specific statistics (one value calculated per unit area)
- Helps identify cluster/hotspots and heterogeneity
- Often referred to as **LISA** (local indicators of spatial autocorrelation)

Local Moran's I (LISA)

- Used to determine if local autocorrelation exists around each areal unit
- Returns an I value for each unit (Global I is simply the average of this)

$$I_i = \frac{(x_i - \bar{x}) \sum_{j=1}^n w_{ij} (x_j - \bar{x})}{\sigma^2}$$

focal area's deviation from the mean

scales the statistic by the overall variability in the data

weighted neighbourhood deviation from the mean

where:

n – the number of spatial objects (the number of points or polygons),

x_i, x_j – are the values of the variable for the compared objects,

$\bar{x}$ – it is the mean value of the variable for all objects,

w_{ij} – elements of a spacial weight matrix (it is recommended that the matrix is row standardized),

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} - \text{variance}$$

Local Moran's I (LISA)

- Used to determine if local autocorrelation exists around each areal unit
- Returns an I value for each unit – Global I is simply the average of this
- Often used after a global Moran's I to investigate:
 - Absolute locations in the study area where clusters of high or low outcome values are occurring
 - Are there local outliers that contribute to a significant global statistic
- Simple Monte Carlo approaches can be used to detect clusters and outliers
 - Cluster of high values (high-high)
 - Cluster of low values (low-low)
 - Outliers (high value surrounded by low values or vice versa)

Example of
published study
using local
Moran's I

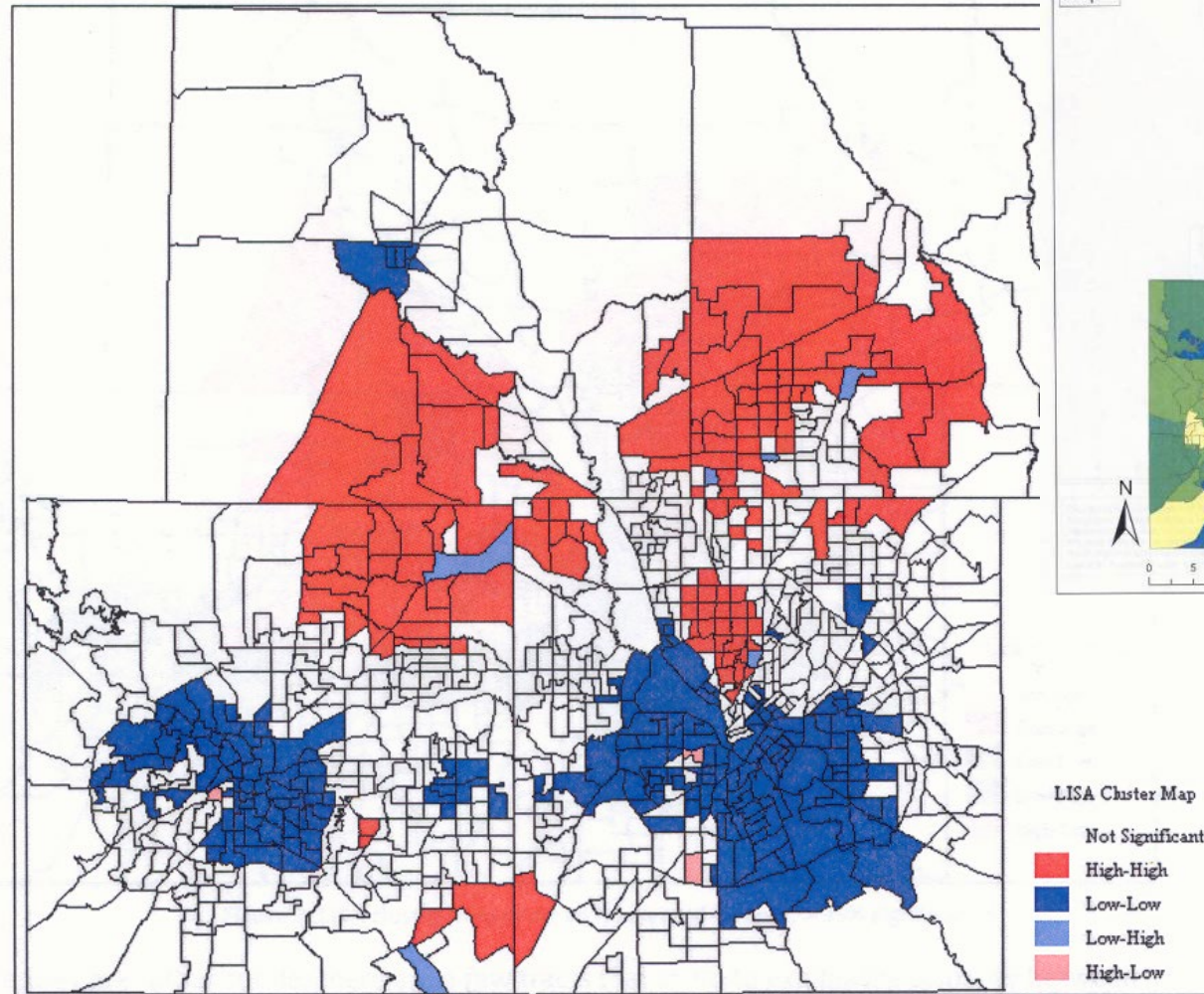
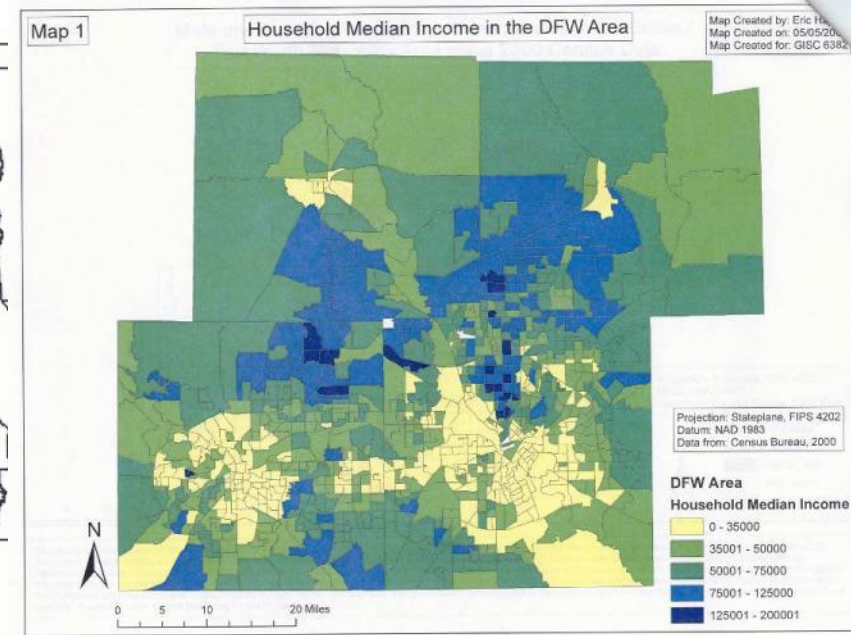


Figure 2: LISA Cluster Map of median income, > 95% significance

LISA for Median Income, 2000 in D/FW

LISA statistic= local indicator of spatial association



Moran's I = .59

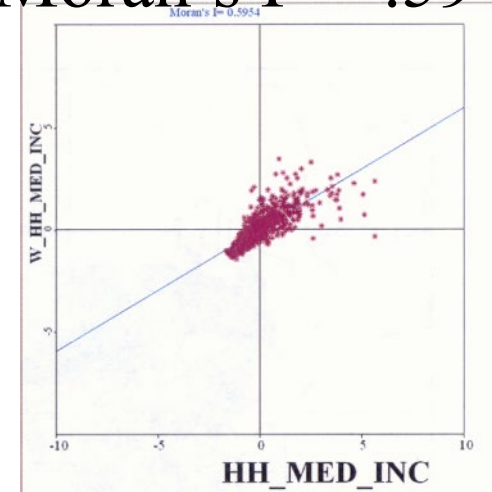
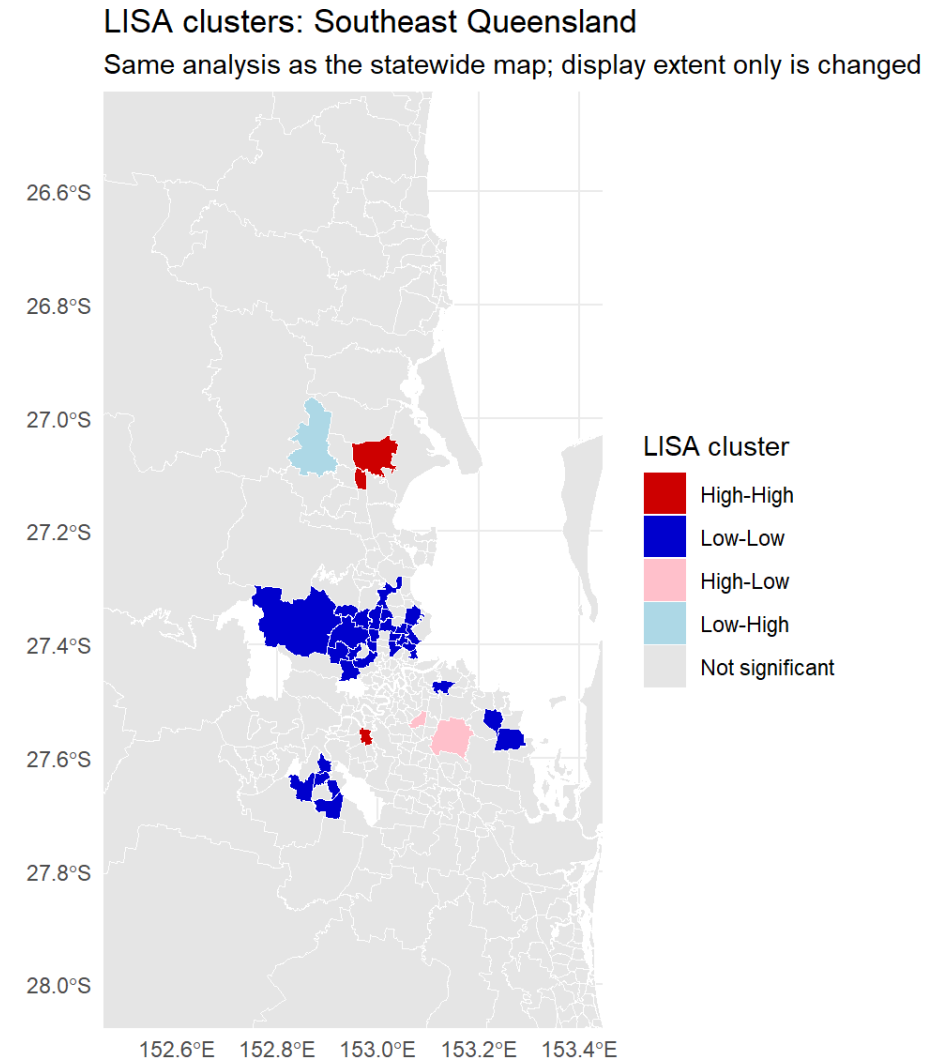
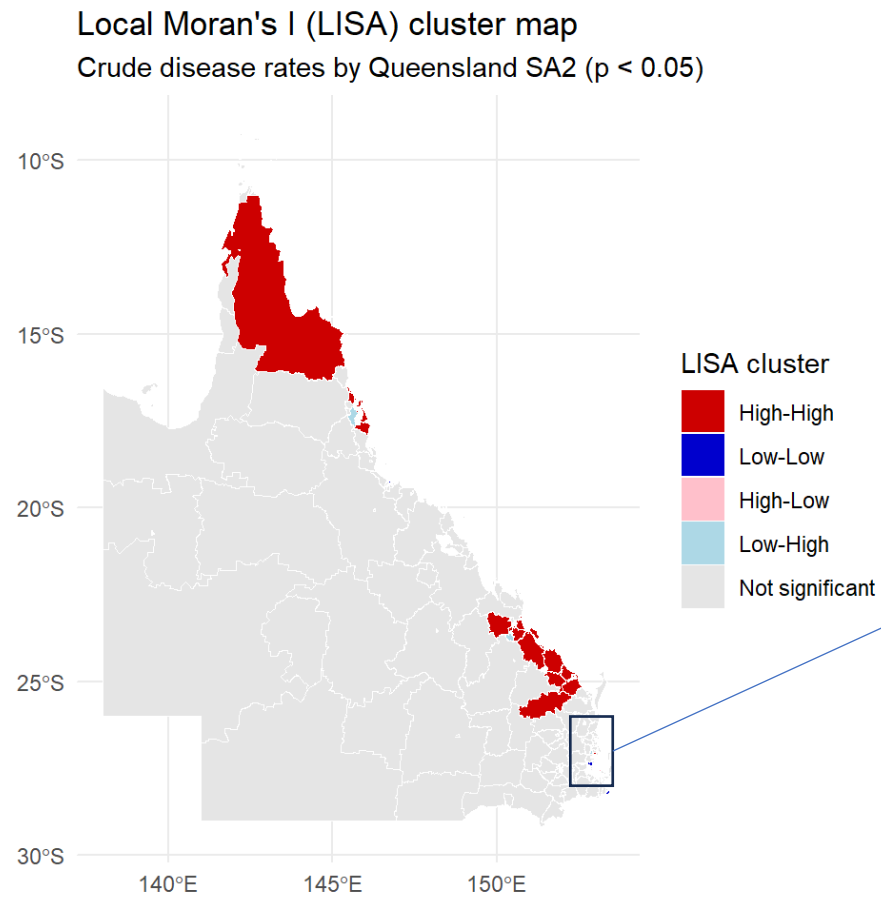


Figure 5: Moran's I Scatterplot of Household Median Income

Example in your tutorial later



To summarise: three main categories of tests to assess spatial clustering/autocorrelation

1. Global: a single test for general patterns and spatial autocorrelation over the entire study region or area
 - Moran's I
 - Geary's C
2. Local: search for specific regions or areas where clustering is observed above or below expected levels
 - Examples: areas with high or low crime rates
 - Local Moran's I
 - **Getis-Ord G_i^***
 - Spatial Scan Statistic (e.g. Kulldorff)
3. Focal: specialised statistics search only in regions around fixed locations
 - Example: cancers around nuclear reactors/power plants
 - Stone's Test
 - Bithell
 - Tango

Getis-Ord local statistic

- **Getis–Ord Local G** is a local spatial autocorrelation statistic used to identify clusters of unusually high or low values (“hot spots” and “cold spots”).
- It compares the **weighted values in each area’s neighbourhood** with the distribution of values across the full study area.
- **G_i** excludes the focal area from its own neighbourhood calculation; **G_i^*** includes the focal area. In practice, G_i^* is most commonly used for hot-spot analysis.
- Results are expressed as **z-scores**:
 - Large positive z-scores: high-value clusters (hot spots)
 - Large negative z-scores: low-value clusters (cold spots)
- In R, calculate local G or G^* using `spdep::localG()`, with self-neighbours included for G^*

$$G_i^* = \frac{\sum_{j=1}^n w_{i,j} x_j - \bar{x} \sum_{j=1}^n w_{i,j}}{\text{Var}(G_i^*)^{1/2}} \quad (4)$$

- $\bar{x}, s$ are sample mean and standard deviation of the target variable; n areas
- the $w_{i,j}$ are the neighbour weights
- the numerator shows the difference between area j ’s weighted average of the target and the overall weighted average
- the denominator standardizes the index
- interpret as Z-score: $+$ $\rightarrow$ clustering of high values, $-$ $\rightarrow$ clustering of low values

Getis-Ord local G – revised formulation

- Generalize [15] to any weighting, not just 0/1 and not just based on distances
- So it can use the same weighting styles as for Moran’s I
- Define as a standard (normal) variate
 - ▶ original G_i less its expectation $W_i = \sum_{j \neq i} w_{ij} / (n - 1) \dots$
 - ▶ ... divided by the square root of the variance:

$$\text{Var}(G_i) = \frac{W_i(n - 1 - W_i)}{(n - 1)^2(n - 2)} \cdot \left[\frac{s(i)}{\bar{x}(i)} \right]^2$$

Local Moran's I vs Getis-Ord G_i^*

- **Both methods assess spatial clustering**, but they emphasise different features of the pattern. Global statistics describe the overall study area, whereas local statistics identify where significant local patterns occur.
- **Getis-Ord General G** asks whether *high values* or *low values* are clustered across the whole area:
 - Positive, significant z-score: clustering of high values.
 - Negative, significant z-score: clustering of low values.
 - Non-significant result: no evidence of overall high- or low-value clustering.
- **Local Moran's I** compares each area with its **neighbours only**. It can identify clusters of similar values—high—high and low—low—as well as spatial outliers—high—low and low—high.[carto+1](#)
- **Getis-Ord G_i^*** includes the area's **own value plus its neighbours** in the local calculation. It is designed specifically to identify:
 - **Hot spots**: local concentrations of high values.
 - **Cold spots**: local concentrations of low values.[pysal+1](#)
- **Key practical distinction**: use **G_i^*** when the question is “Where are the hot and cold spots?” Use **Local Moran's I (LISA)** when the question also includes “Where are unusual spatial outliers?”

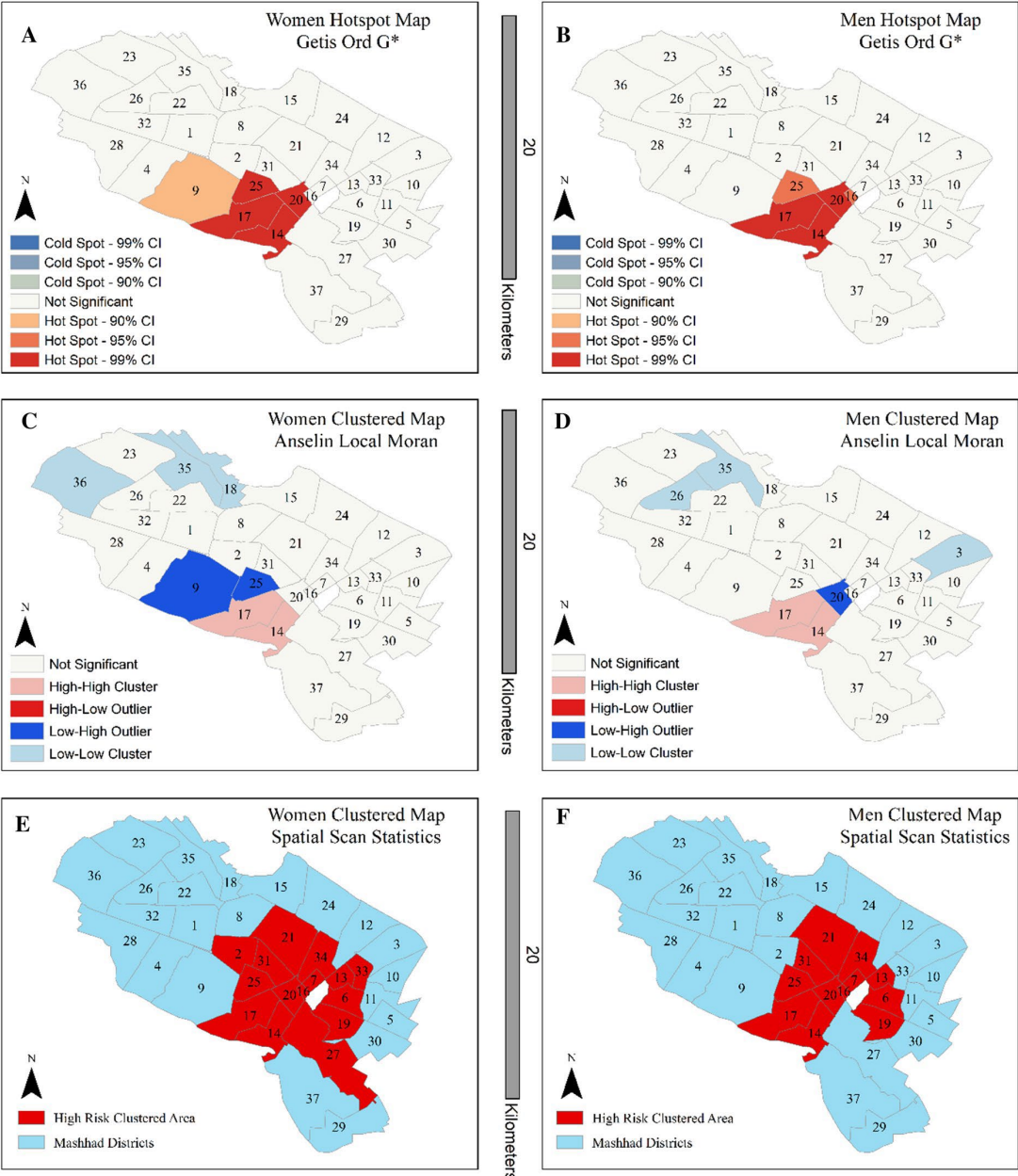
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Alternative methods for local cluster analysis: spatial scan statistics

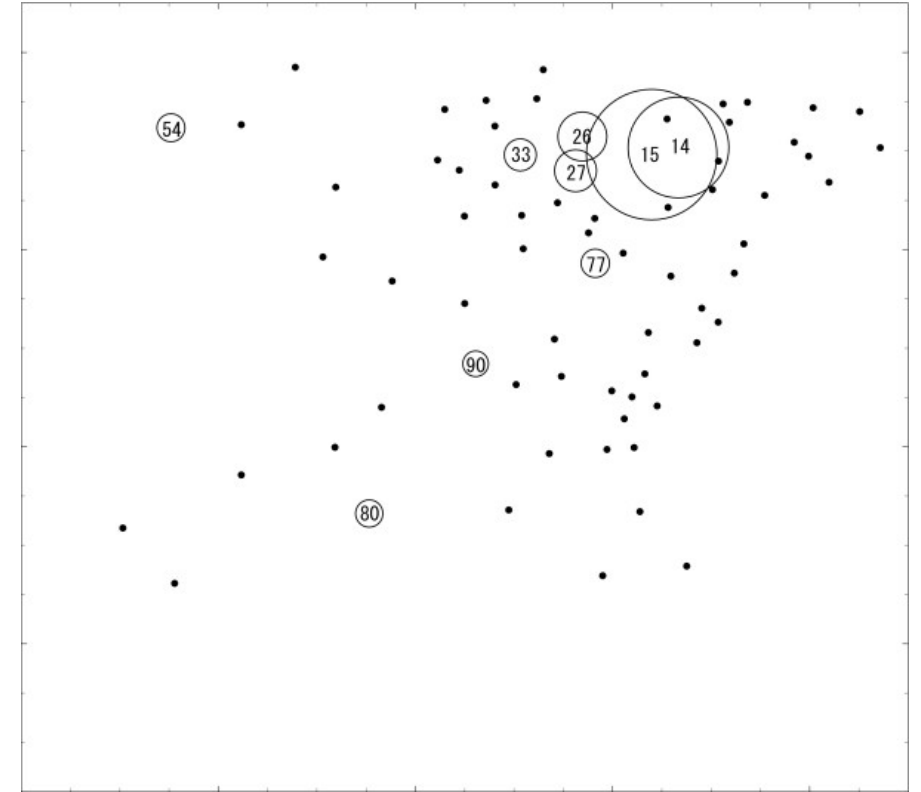
- These are based on comparisons of local rate estimates
 - Test whether the ratio of cases to controls (or cases per population at risk) appear “significantly” elevated in certain areas
 - Goal is to detect specific clusters where the observed rate (or ratio) is inconsistent with the rate (or ratio) over the rest of the study area.
- In contrast to a global statistic, this method is used to identify the ***location*** of clusters.
- Global and local analyses can give conflicting results → Statistically significant clusters can be identified within a dataset which, overall, displays a CSR pattern.
- Identify spatial, temporal or spatial-temporal clusters; controls for risk factors or covariates (e.g. confounding)

Getis Ord vs Local Moran vs Spatial Scan Statistics



Kulldorff spatial (and spatial-temporal) scan statistics

- Developed by Martin Kulldorff
- Detects clusters in space and/or time
- Implemented in SaTScan software



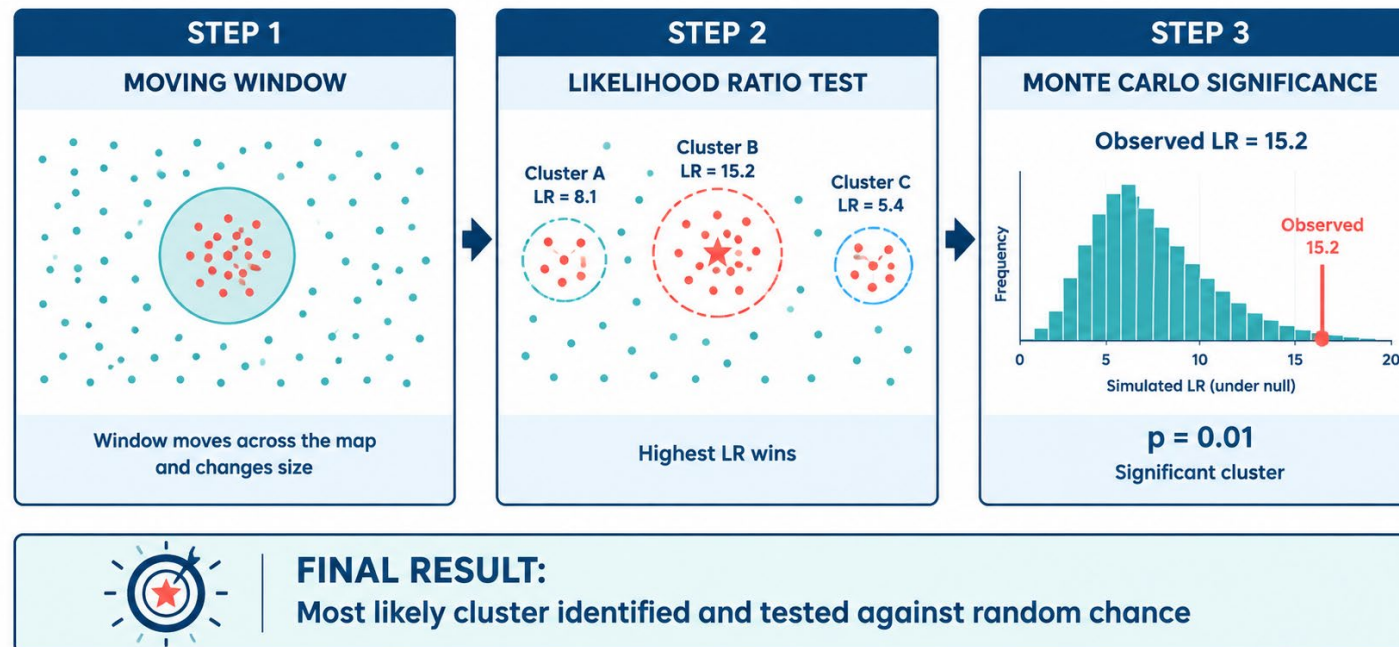
<https://pmc.ncbi.nlm.nih.gov/articles/PMC1173134/>

SaTScan – Logic & Method

- Moving window scans
- Likelihood ratio test
- Monte Carlo simulations to assess significance

KULLDORFF SPATIAL SCAN STATISTIC: HOW IT WORKS

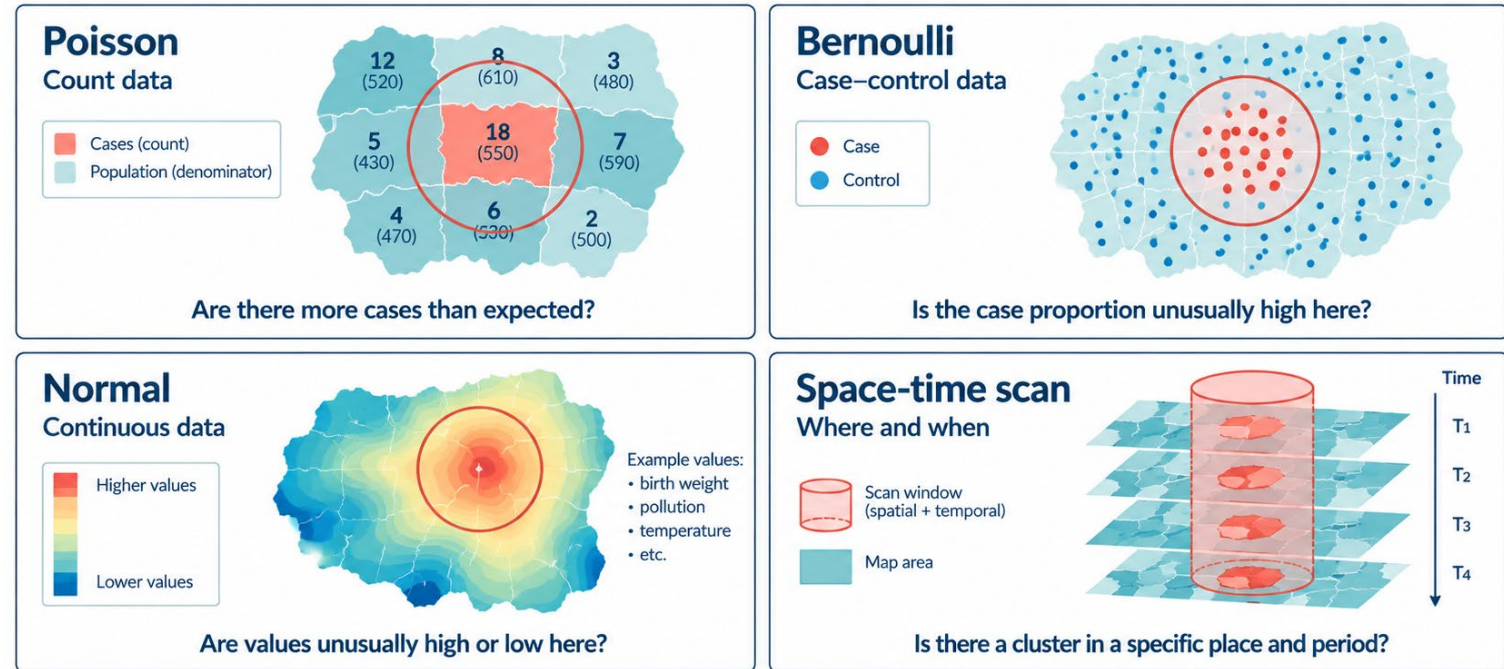
Detecting statistically significant clusters



SaTScan – Model Types

- Poisson (count data)
- Bernoulli (case-control)
- Normal (continuous)
- Space-time scan

KULLDORFF SCAN STATISTICS: COMMON DATA MODELS



Each model scans candidate windows and tests whether the observed pattern is unlikely under random chance.

Graphic created using Perplexity.ai

SaTScan – public health applications

- Early outbreak detection and investigation.
- Targeting testing, vaccination, vector control, and health promotion.
- Monitoring inequities and geographic disease burden.
- Supporting environmental-exposure and cancer-cluster investigations.

Analysis	Main question	Example use
Spatial scan	Where are unusually high disease rates?	Identify high-burden areas for TB, malaria, cancer, or vaccination outreach.
Time scan	When did cases increase unexpectedly?	Detect a sudden rise in gastroenteritis presentations.
Space–time scan	Where and when is an outbreak occurring?	Detect a local outbreak in time of salmonellosis, legionellosis, shigellosis, or COVID-19.

Spatiotemporal clustering of COVID-19 mortality rates



*Tropical Medicine and
Infectious Disease*



Article

The COVID-19 Mortality Rate Is Associated with Illiteracy, Age, and Air Pollution in Urban Neighborhoods: A Spatiotemporal Cross-Sectional Analysis

Alireza Mohammadi ^{1,*}, Elahe Pishgar ², Munazza Fatima ^{3,4}, Aynaz Lotfata ⁵, Zohreh Fanni ², Robert Bergquist ^{6,†} and Behzad Kiani ⁷

¹ Department of Geography and Urban Planning, Faculty of Social Sciences, University of Mohaghegh Ardabili, Ardabil 56199-11367, Iran

² Department of Human Geography, Faculty of Earth Sciences, Shahid Beheshti University, Tehran 19839-69411, Iran

³ Department of Geography, The Islamia University of Bahawalpur, Bahawalpur 63100, Pakistan

⁴ Department of Geography, University of Zurich, CH-8006 Zurich, Switzerland

⁵ Geography Department, Chicago State University, Chicago, IL 60628-1598, USA

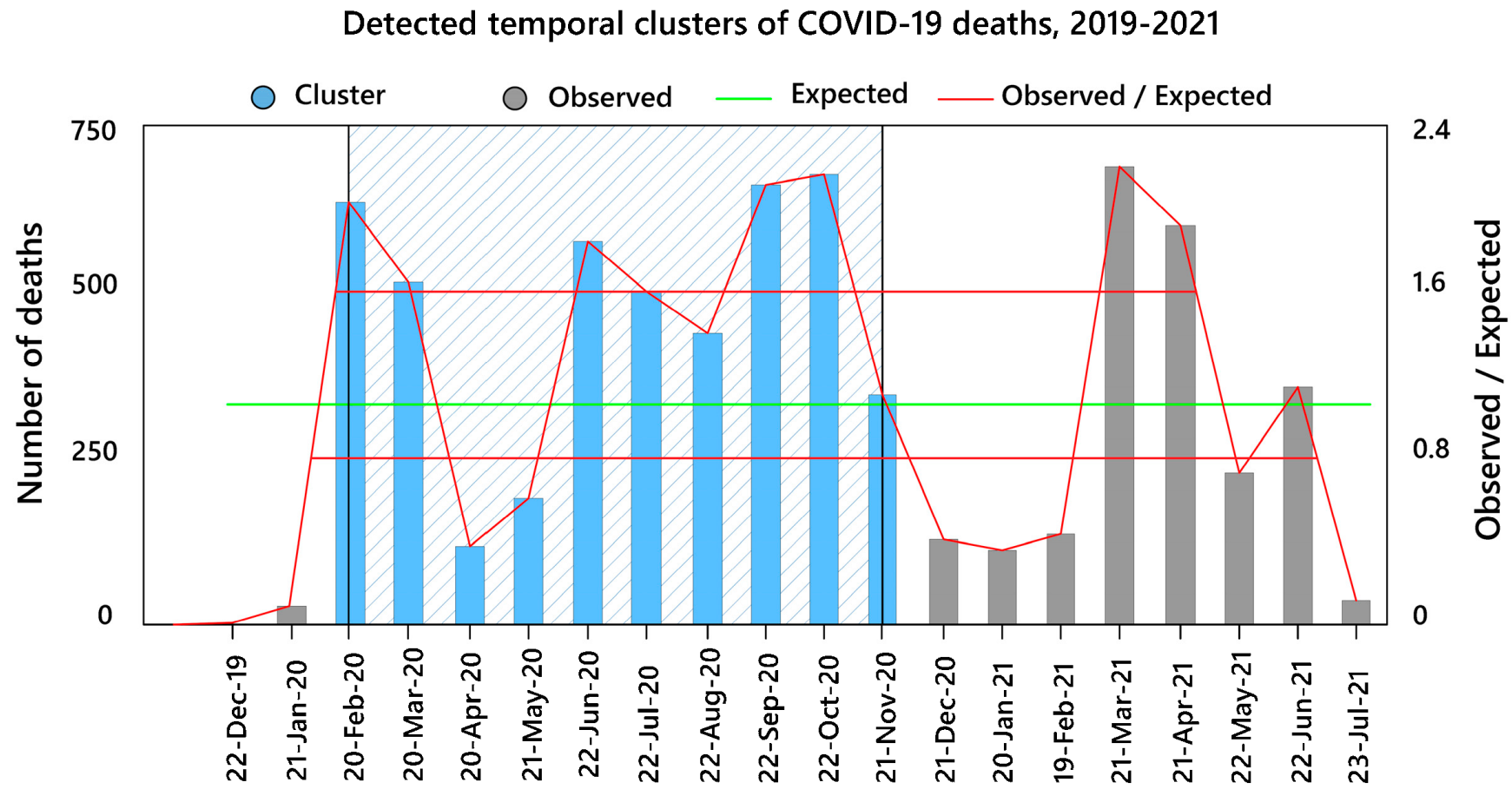
⁶ Ingerod, SE-454 94 Brastad, Sweden

⁷ Centre de Recherche en Santé Publique, Université de Montréal, 7101, Avenue du Parc, Montreal, QC H3N 1X9, Canada

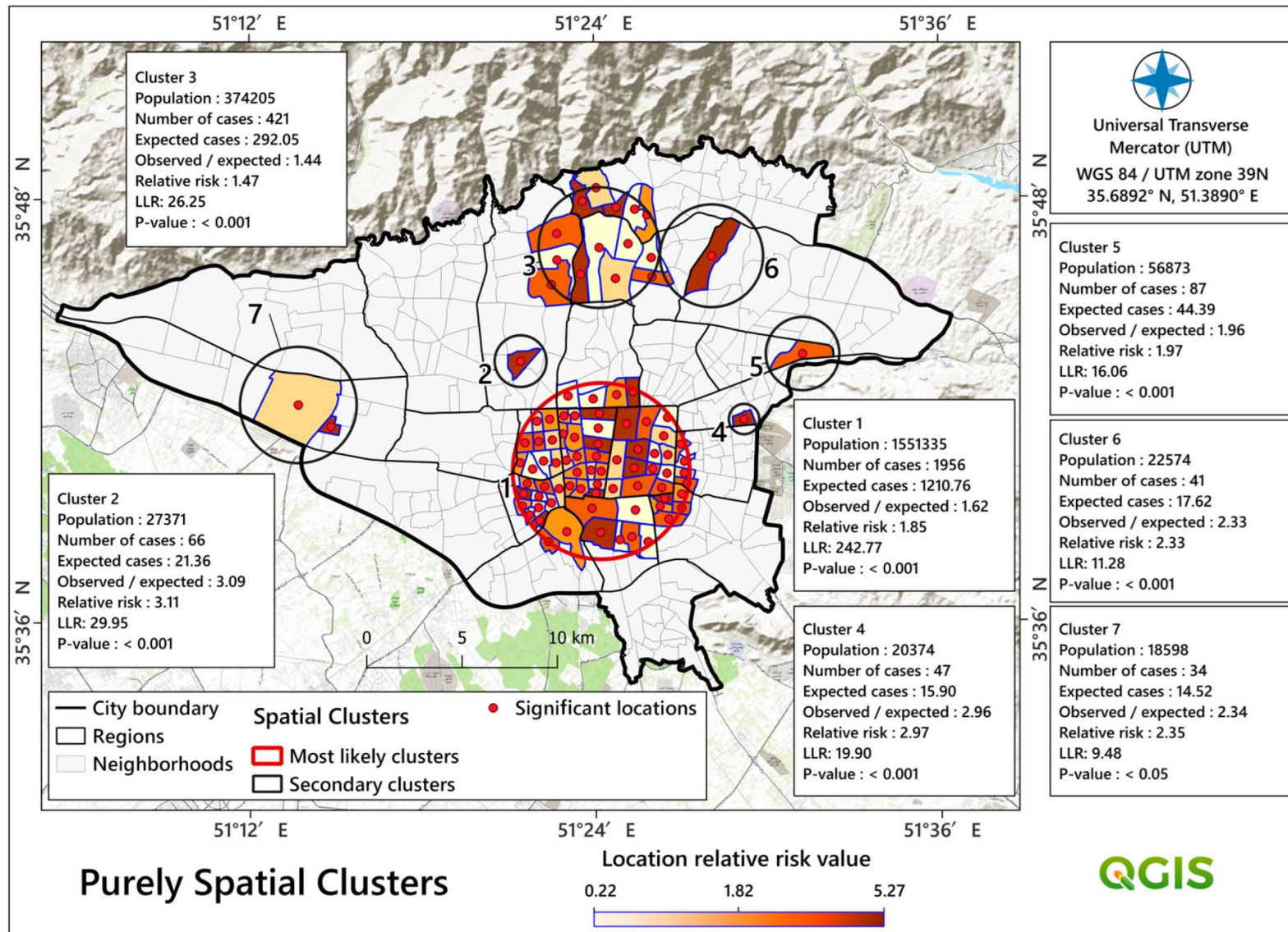
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† Formerly UNICEF/UNDP/World Bank/WHO Special Programme for (Re-search and Training in Tropical Diseases (TDR), World Health Organization, Geneva, Switzerland.

Purely temporal clusters



Purely spatial clusters (1)



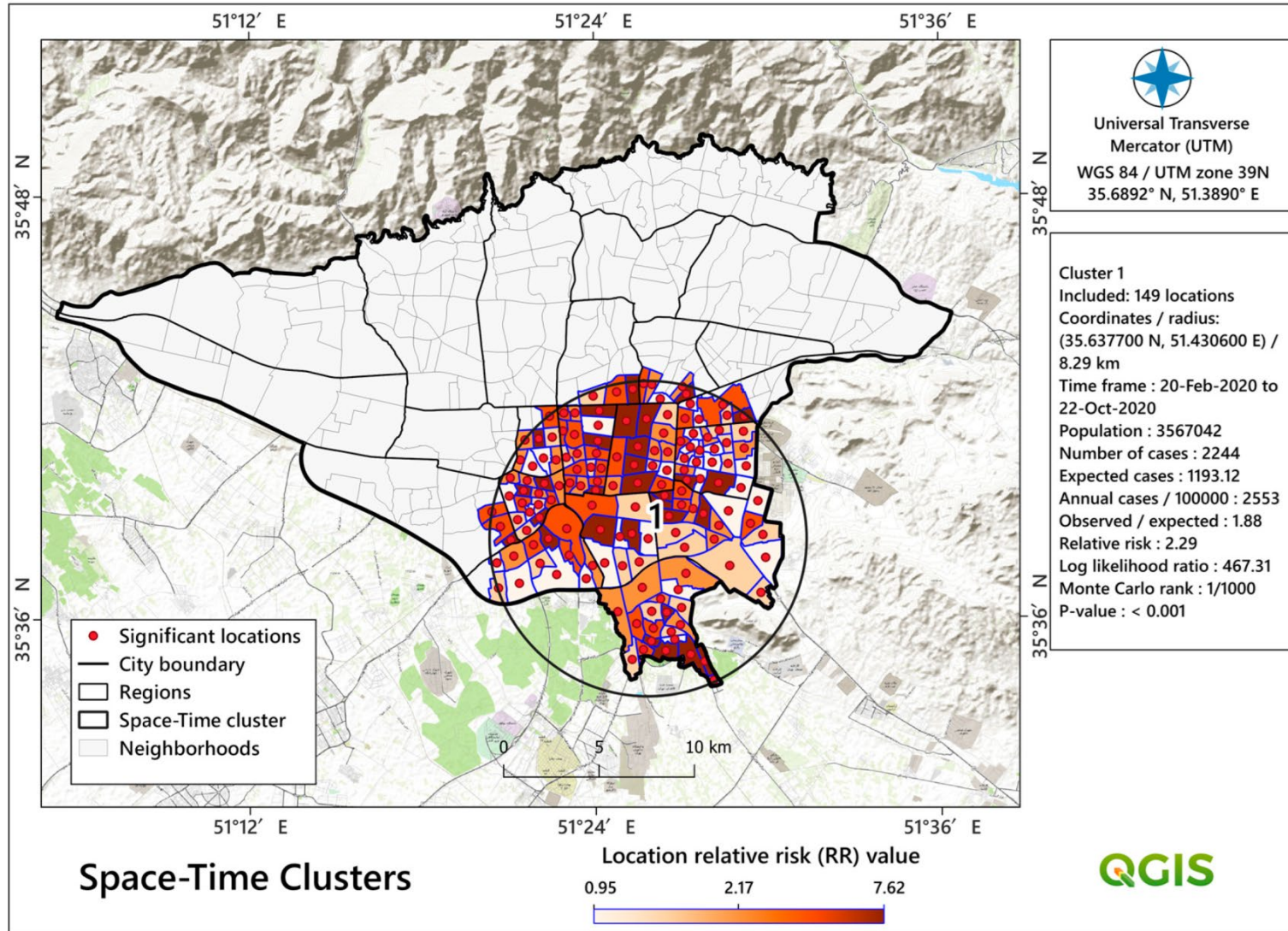
Purely spatial clusters (2)

Table 2. Purely spatial detected clusters based on a Poisson scan statistic model for areas with high rates of COVID-19 deaths in Tehran, Iran.

Cluster	Neighbourhood	Coordinates/Radius	Population	Cases (No.)	Expected Cases (No.)	Annual Cases/100,000 (No.)	Observed/Expected Cases (No.)	Relative Risk	Log Likelihood Ratio	Monte Carlo Rank	p-Value
1	Includes seventy-one neighbourhoods	(35.671100° N, 51.403600° E)/4.63 km	1,551,335	1956	1210.76	2192.9	1.62	1.85	242.78	1/1000	<0.001
2	Includes one neighbourhood	(35.723200° N, 51.357100° E)/1.3 km	27,371	66	21.36	4193.9	3.09	3.11	29.95	1/1000	<0.001
3	Includes fourteen neighbourhoods	(35.776700° N, 51.402900° E)/3.17 km	374,205	421	292.05	1956.7	1.44	1.47	26.25	1/1000	<0.001
4	Includes one neighbourhood	(35.695400° N, 51.486500° E)/0.8 km	20,374	47	15.90	4012.2	2.96	2.97	19.9	1/1000	<0.001
5	Includes one neighbourhood	(35.726200° N, 51.520600° E)/1.87 km	56,873	87	44.39	2660.6	1.96	1.97	16.06	1/1000	<0.001
6	Includes one neighbourhood	(35.772600° N, 51.468100° E)/2.6 km	22,574	41	17.62	3158.9	2.33	2.33	11.28	3/1000	<0.001
7	Includes two neighbourhoods	(35.702800° N, 51.228200° E)/2.05 km	18,598	34	14.52	3179.6	2.34	2.35	9.48	25/1000	<0.001

A cluster is statistically significant when its log likelihood ratio is greater than the critical value, which is, for the significance level: Gumbel Critical Values: 0.00001: 18.16;... 0.0001: 15.57; ... 0.01: 5.21. Standard Monte Carlo Critical Values: 0.001: 12.545047; 0.01: 10.568609; 0.05: 8.628534.

Spatiotemporal clusters



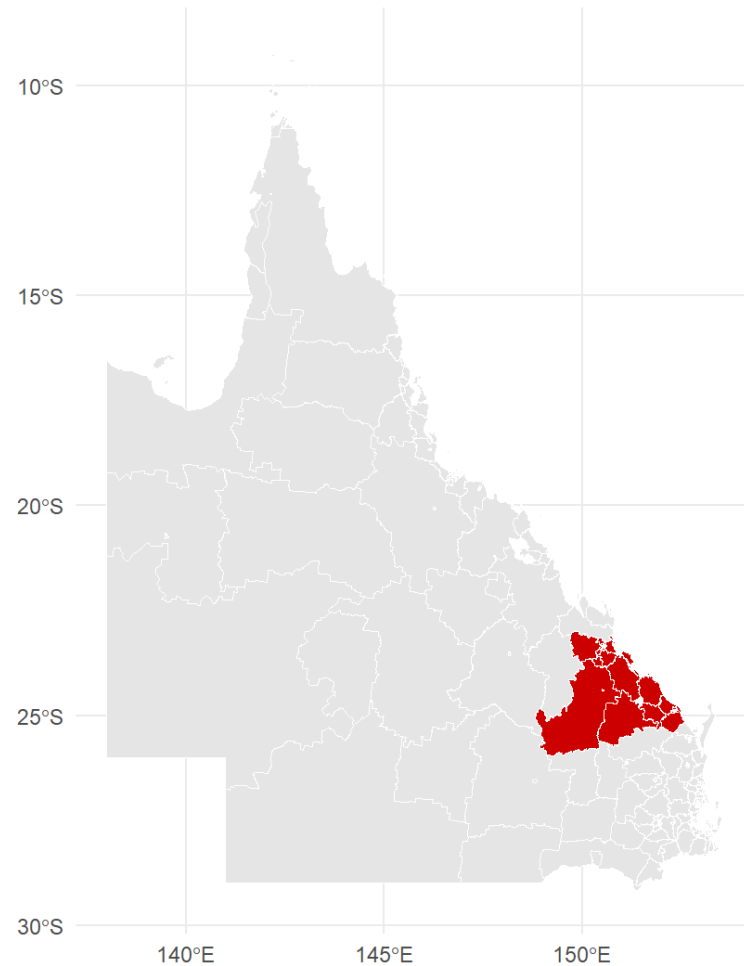
Preparing Data for SaTScan (Poisson Model): Queensland SA2 Example

- **Case file**: Number of cases per location (e.g., per SA2). Time variables (if doing space-time scan).
- **Population file**: Denominator population at risk per location
- **Coordinates file**: Centroids (lat/long) of each location

Example in your tutorial

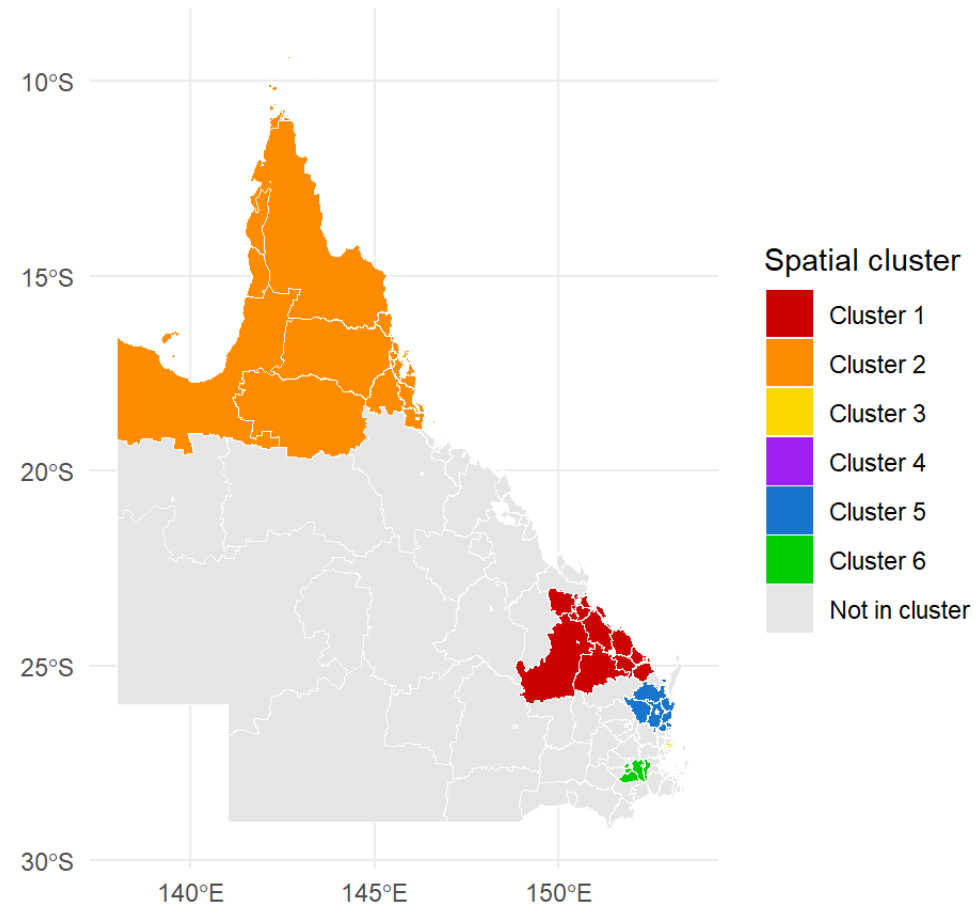
Most likely Kulldorff spatial cluster

Poisson spatial scan statistic, maximum population window = 10%



Kulldorff spatial clusters of disease risk

Significant clusters from the Poisson spatial scan statistic



So what? Identifying potential explanations

- A lot of exploratory spatial analyses of local autocorrelation stop at identifying clusters
- Perhaps a more important question is – why are they there?

We can look at this by:

- Using covariate data to explore what factors are associated with ‘being a cluster’ – e.g. regression model
- Bivariate Moran’s I (ie the correlation between one variable and the spatial lag of another variable)
- Spatially-explicit discrete modelling frameworks (conditional autoregressive) – [we will get into this during session.](#)

Software & Tools

- ArcGIS
- GeoDa
- R packages: spdep, sf, tmap, scanstatistics
- SaTScan software

R tutorial

- <https://behzadkiani.github.io/spatial-epidemiology-r-tutorial/>
- Create two plausible spatial-weights matrices for the same areas (maybe Queen contiguity vs distance-based threshold e.g. K-Nearest Neighbors (KNN) within 100km?)
- Calculate global Moran's for rates and interpret the result in plain language
- Create a local cluster/hotspot (e.g. local Moran's I or LISA) map
- Compare outputs under alternative neighbour definitions
- Calculate Kulldorff space-time cluster statistic for rates (high and low) to highlight extension from spatial only clustering to space-time clustering and interpret the result in plain language